

DIGITAL PRODUCT DESIGN

Lecture 7: Choosing Metrics

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Outline

- Probability in 1 Slide
- Organizational metrics
- Combining Metrics
- Targeting for Long Term Outcomes
- Engagement as a Metric

PROBABILITY IN 1 SLIDE

Probability Review

- Random variable:
 - A variable whose value cannot be predicted exactly
- Expected value ($E[X]$):
 - Average X in the population
- Conditional probability ($\Pr(Y = y|X = x)$):
 - An event's likelihood ($Y = y$) given the realization of another random variable ($X = x$)
- Conditional expectation $E[Y|X = x]$:
 - Average Y in the subpopulation where $X = x$

ORGANIZATIONAL METRICS

KOHAVI, TANG, AND XU (2020)

What Are Metrics?

- Metrics are what you measure
- Good metrics proxy the long-term objective of the organization

Types of Metrics

- **Goal metrics:** Reflect what organization ultimately cares about
- **Driver metrics:** Short-term and more sensitive than goal metrics
 - Aka sign post metrics, surrogates, indirect
 - Reflect a causal model of what moves the goal metrics
- **Guardrail metrics:** Verify assumptions and protect organization
 - Examples include user-experience (page load) or trust / safety stats

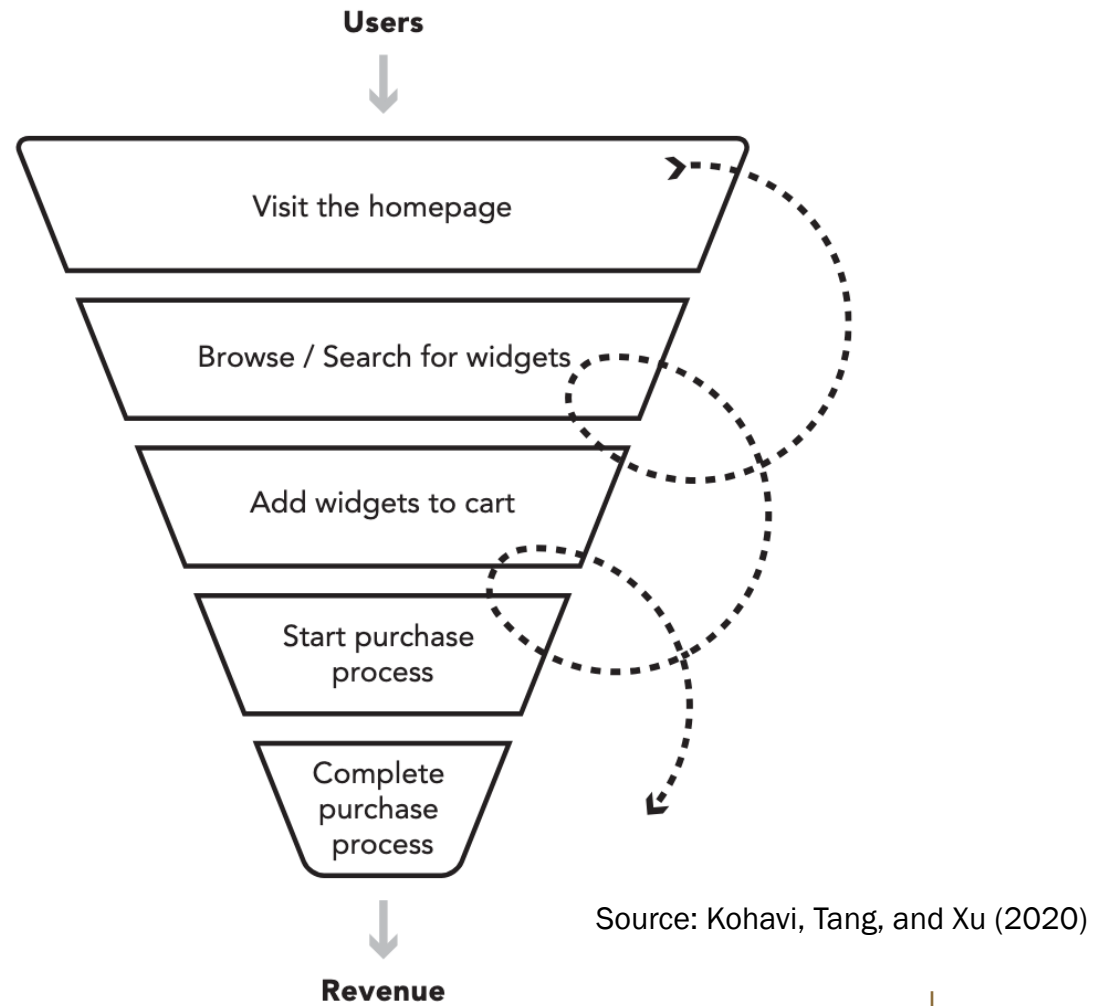
Goal Metrics

- **Goal metrics:** Reflect what organization ultimately cares about
- Directly related to organization's mission
- Often hard to measure in the short-run
- Helpful to articulate organization's mission in words and then proxy this goal with metrics
 - Proxy is often imperfect, constantly refine
 - Important to understand metrics are a proxy for this goal

Driver Metrics

- **Driver metrics:** Short-term and more sensitive than goal metrics
 - Aka sign post metrics, surrogates, indirect
- Reflect a causal model of what moves the goal metrics
- Often includes “up-the-funnel” metrics

Driver Metrics, Part 2



Driver Metrics, Part 3

- Different teams may have different driver metrics
- Example: Content business adopting the advertising revenue model wants to maximize long-term profits.
 - What is a good driver metric for the content team?
 - What is a good driver metric for the support team?

Guardrail Metrics

- **Guardrail metrics:** Verify assumptions and protect organization
 - User-experience (page load time)
 - Trust / safety stats
 - Security often in conflict with ease of use

Additional Metrics

- Useful to collect a suite of metrics related to key metrics
 - E.g. If CTR is a key metric, we might want to track where clicks happen on the page to understand what is moving the CTR

Question: If revenue is a key metric, what other metrics might we want to track?

- Extensive / intensive margin, category of sales, referral source

Challenges of Metrics

- Goal metrics:
 - Long-run and hard to move
- Driver metrics:
 - Potentially distracting
 - Difficult to find good driver metrics
 - Potentially gameable
- Guardrail metrics:
 - Easy to look past

Defining Metrics

- Measure relationships between data you collect and goal metrics
- Example: One of Netflix's goal metrics is the retention rate of existing customers. This is a slow-moving target, what are possible driver metrics?
 - Streaming hours is correlated with retention
 - Relationship is non-linear
 - Most important to move light users to moderate users than heavy users to heavier users (Xie and Aurisset, 2016)

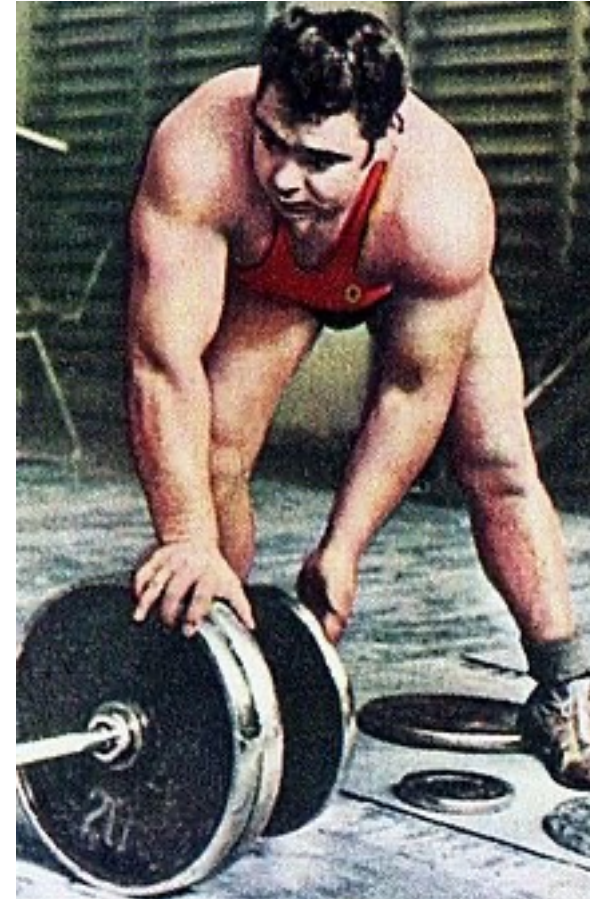
Defining Metrics, Continued

- Measure relationships between data you collect and goal metrics
- Example: Facebook has a goal of showing content users like. How can we find driver metrics for this goal?
 - Rank content by predicted engagement
 - Assumption is that people engage with what they like
 - Survey users: “Is this post worth your time?” (Gupta, 2021)
 - Is this scalable?
 - One way to scale: survey a subset of users and predict survey responses

Gameability of Metrics – Vasily Alekseyev

- Soviet government paid a bonus every time he broke a WR
- What do you think happened?
- He broke his own record ~80 times increasing WR by 1lb each time

Source: Spitzer (2007), Kohavi, Tang, and Xu, (2020)



Gameability of Metrics – The Chicken KPI

- Fried chicken chain evaluated managers based on the Chicken Efficiency Index (Chicken Sold / Chicken Cooked)
- *“Oh yeah we get tons of kids coming in here after baseball or soccer practices or games, families, and lots of people. I tell them I’m going to make their chicken to order and it will take 15-20 minutes.”*
- *“they file right out the door and head off someplace else, but management doesn’t measure that.”*

Source: Spitzer (2007), Brown (2019), Kohavi, Tang, and Xu, (2020)

Gameability of Metrics – Hanoi Rat Bounty

- There was a program in Hanoi paying a bounty for every rat tail handed in to control the rat population
- Led to the growth of a small industry of breeding and raising rats

Source: Wikipedia contributors, Kohavi, Tang, and Xu, (2020)



Source: ABC

Takeaways

- Finding good metrics is hard
 - Start with the organization's mission, proxy this with goal metrics
 - Proxy goal metrics with short-term driver metrics
 - Set up effective guardrail metrics to protect the business
- Avoid gameable metrics
 - “Tell me how you measure me, and I will tell you how I will behave”
 - Goldratt (1990)
 - “Show me the incentives, and I'll show you the outcome”
 - Charlie Munger (1995)

OVERALL EVALUATION CRITERIA

KOHAVI, TANG, AND XU (2020)

Metrics Useful for Decisions

- Data-driven organizations have goal, driver, and guardrail metrics to achieve their goals and align their efforts
- These are not always useful for decision making
- Metrics useful in decision making must be
 - Measurable in a scalable way
 - Sensitive & timely
 - Stock price is insensitive to most decisions
 - Feature availability is sensitive

Metrics Useful for Decisions - Examples

- User experience: Hard to measure user happiness
 - Proxy with behavioral data
- Ad-revenue: Common for outliers with high CPC to affect revenue
 - CTR or truncated revenue could be more sensitive
- Subscription: Metric moves with the renewal cycle
 - Perhaps shorter-term surrogates are more sensitive

Combining Key Metrics into an OEC

- Organization has many metrics they care about
 - There is often a mental model of tradeoffs among these metrics
 - What drop in active users is acceptable for an increase rev per user?
- Combining these metrics into an OEC (Overall Evaluation Criteria) allows you to codify tradeoffs and quantitatively define the objective
 - Often (but not necessarily) a weighted average of metrics
- Aligns people in the org about tradeoff and what constitutes success
 - Results in more consistent decisions

Deciding on the OEC

- This is very context dependent
 - Should proxy the highest goal metrics
- This can (should) change over time
 - If the decision that maximizes the expected OEC seems wrong, revisit the OEC. What is missing?

OEC Case Study – Amazon Emails

- Amazon evaluating a program to send emails to customers recommending new books
 - Initial OEC was short-term revenue
- More emails => Weakly more short-term revenue
- Complaints started to come in about spam emails
 - Annoyed users unsubscribe, lose potential for future emails

OEC Case Study – Amazon Emails, Continued

- Refine the OEC:

$$OEC = E[Revenue] - L \times Pr(Unsub)$$

- Revenue: Revenue per user
 - Unsub: Whether a user unsubscribes
 - L: Dollar value of lost revenue due to unsubscribe
-
- When implemented with even a small L , more than half of email campaigns were determined to be harmful

Source: Kohavi, Tang, and Xu (2020)

OEC Case Study – Bing Search

- Bing introduced a bug that led to bad organic search results
- Two metrics shot up: queries per user and revenue per user
 - Why?
- Bing had to revise their OEC to include driver metrics that better proxy their goal metrics
 - Sessions per user became more important, as satisfied users are likely to return

Source: Kohavi, Tang, and Xu (2020)

BOSTON GLOBE

YANG, ECKLES, DHILLON, AND ARAL (2023)

- Newspapers increasingly turning to subscriptions for primary revenue source
- Goal metric of long-term customer value (retention of existing customers)
- Deciding how to target marketing interventions
 - Price discounts in this setting
- Key problem: Long-term customer value is realized in long-term!
- Solution 1: Try a policy, wait, update actions once you observe the outcome
- Solution 2: Proxy goal metric with short-term driver metric and iterate faster

Methodology

- Estimate the long-term outcome in historical dataset using data observed in short-run
 - Estimate a conditional expectation: $E[Y|X]$. What is Y ? What is X ?
 - Key assumption: This represents a causal relationship (moving X actually affects Y)
 - Important features:
 - Recent engagement, CC expiration

Methodology, Continued

- Choose discount for each consumer to maximize predicted retention: $E[Y|X]$
 - I.e. the discount that maximizes expected long-term revenue

The Boston Globe

Thank you.

Dear Jessie,

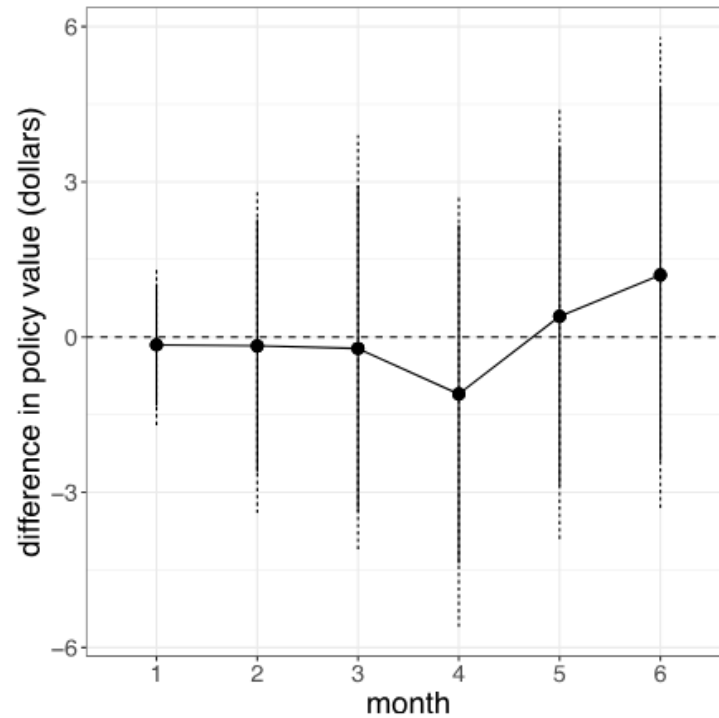
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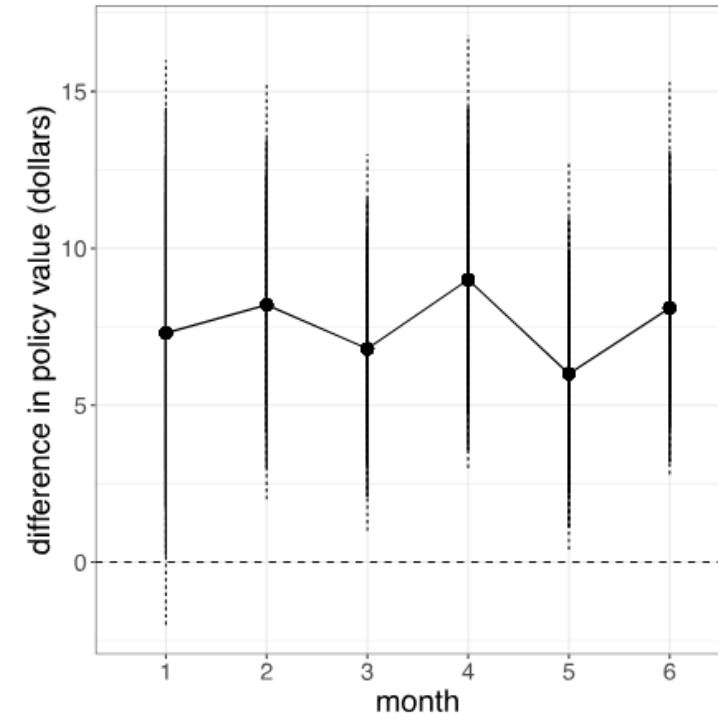
[Go to Globe.com →](#)

Results

Policy Targeting Short-Term Revenue



Policy Targeting Expected Long-Term Revenue



Optimizing for LT proxies increased revenue by ~\$5 million

Recap

- Targeting the right metrics are important
 - Can have real impact on bottom line
- Pick driver metrics carefully
 - They should be on the causal path to ultimate goal metrics
 - Use data to find right driver metrics & combine into an OEC
 - Question: What is the OEC in this case?
- We can then use driver metrics / OEC to guide decisions more effectively

ENGAGEMENT AS A METRIC

Why Engagement as a Metric

- Engagement often defined as clicks, likes, time spent, comments ...
- Many digital products measure success through engagement
 - In part due to the dominance of advertising revenue model
- Engagement signals are easy to collect and highly sensitive

Why Engagement as a Metric, Continued

- Engagement is related to customer value
 - Customers engage with things they like to engage with!
- However, focus only on engagement can cause issues

Concerns with Optimizing for Engagement

- Could the user get the same value faster?
 - Does adding interstitial pages increase time? User value?
- Does this incentivize temptation goods?
 - I eat more Cheetos than I'd like, I watch more YouTube than I like
- Many human biases exist where our actions differ from our “true” preferences
 - How deliberative are you when you are scrolling through social media?
 - Unconscious bias when we don't deliberate
- Have you ever tried to “train” your social media algorithm?

Concerns with Optimizing for Engagement, Continued

“One of the biggest issues social networks face is that, when left unchecked, people will engage disproportionately with more sensationalist and provocative content.

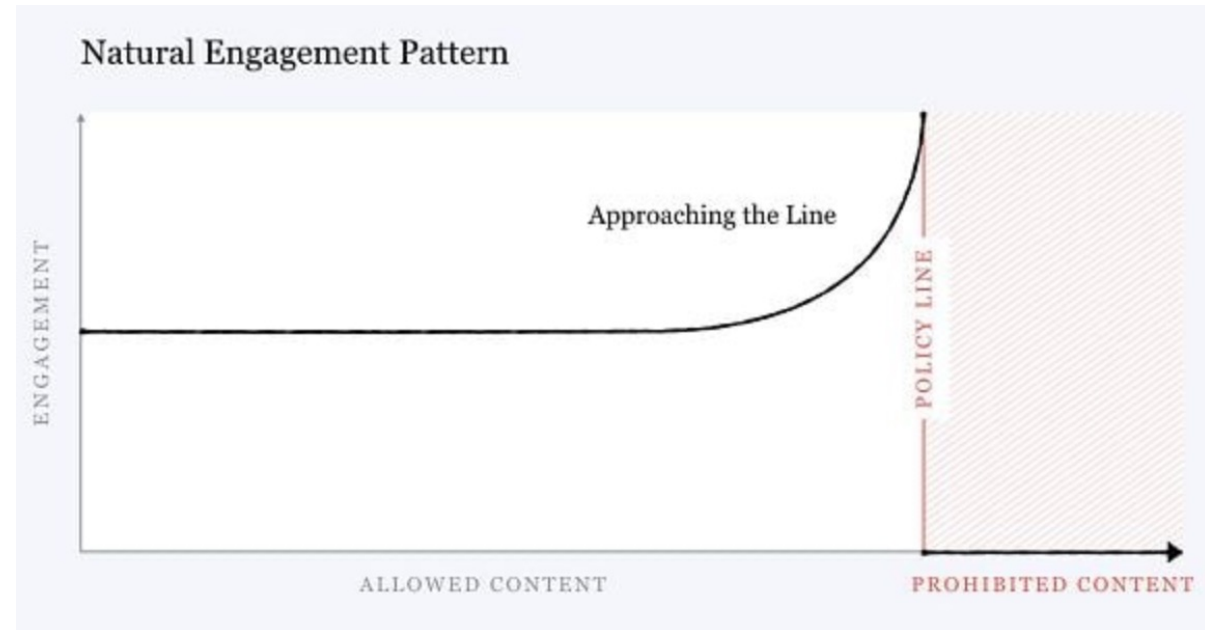
...

People consistently tell us these types of content make our services worse – even though they engage with them.”

— Mark Zuckerberg, 2018

Additional Concerns – Externalities Among Users

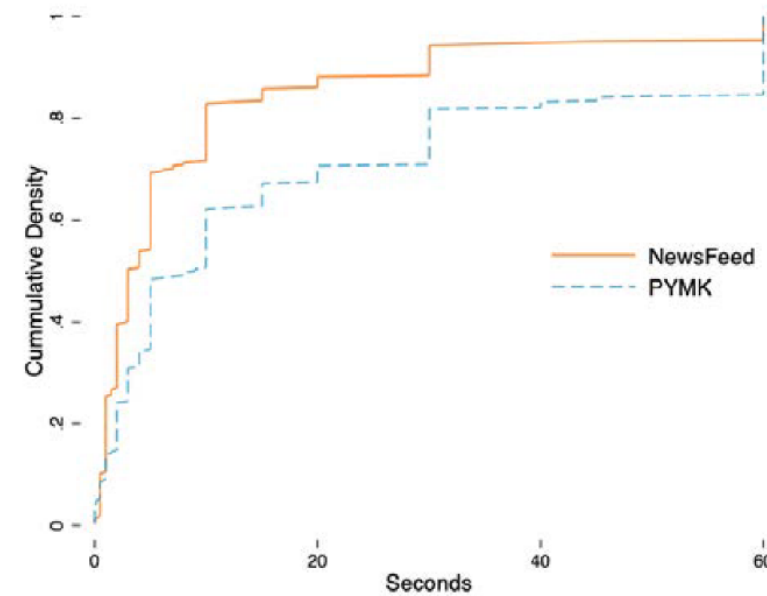
- Are there externalities among users?
- Certain types of content may be highly engaging to some users, but drive others away



Source: Zuckerberg (2018)

Exacerbating Biases

- Study comparing FB people you may know algorithm (PYMK) and the news feed
- PYMK engagement decisions take much longer than news feed decisions
- Evidence of out-group bias on news feed (e.g. White people less likely to see posts from Black friends) but not on PYMK



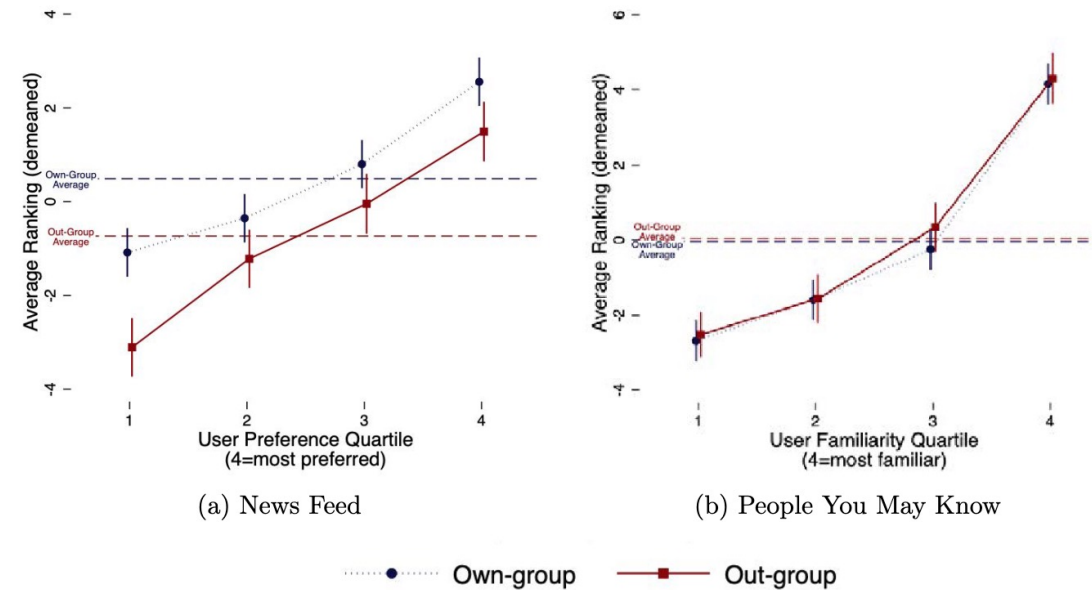
(b) Speed (time to decide in seconds) CDF

Source: Agan, Davenport, Ludwig & Mullainathan (2023)

Exacerbating Biases, Continued

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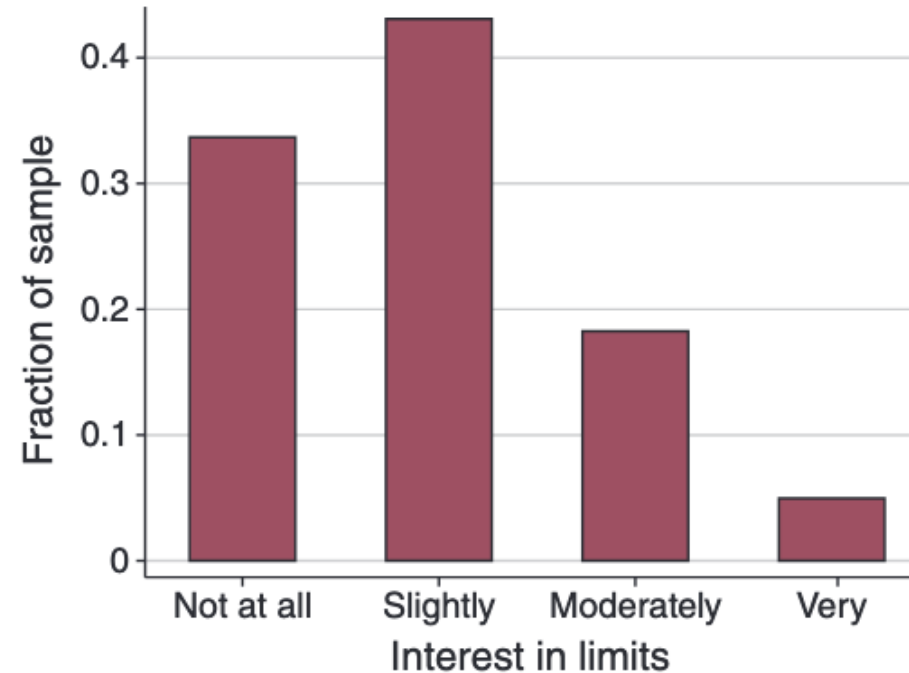
Figure 4: Relationship between News Feed Algorithmic Ranking and Own-group Status Conditional on Subject Explicit Preference



Source: Agan, Davenport, Ludwig & Mullainathan (2023)

Additional Concerns – Short vs Long Term Engagement

- Many people express an interest in technology that limits their access to engagement-based services
- This does not happen with standard goods
- Many people take breaks from social media, requires uninstalling the app / deleting you account



Source: Allcott, Gentzkow, and Song

SUMMARY

Engagement as a Metric Recap

- Goal metrics proxy the mission
- Driver metrics proxy the goal
- Guardrail metrics protect the organization
- Overall Evaluation Criteria encodes tradeoffs between metrics into an objective function
- Careful selection of driver metrics has big payoffs (e.g. Boston Globe)
- Engagement often a useful metric